

posed max block size (PMBS).

Length (LEN)

NOTE This field is only present in **ReceiveAccept** messages, and only if **RC[DEFLen] = 1**.

The Length (LEN) field specifies the length of the transfer. If the Initiator indicated a definite length, this length SHALL either be:

- equal to the proposed definite length, if the remaining data in the file beyond the Start Offset (STARTOFS) is larger or equal to the proposed length;
- smaller than the proposed definite length, if the remaining data in the file beyond the Start Offset (STARTOFS) is smaller than the proposed length.

If this field is present, and the Initiator indicated an indefinite length (i.e. **RC[DEFLen] = 0** in **ReceiveInit**), this Length field SHALL be equal to the size of the file remaining (if known), or 0, for indefinite. The absence of the Length (LEN) field implies indefinite length.

Metadata (MDATA)

The Metadata (MDATA) field is optional and allows the Responder to provide application-specific metadata about the transfer in **TLV format**. The TLV metadata consumes the rest of the payload for the **SendAccept/ReceiveAccept** message, after all previous fields.

11.22.6. Data Transfer Messages

11.22.6.1. Block Ordering Rules

The following behavior applies to all messages containing a Block Counter field.

Queries (**BlockQuery** message) SHALL be made in ascending and sequential Block Counter order. If the arriving Block Counter at the recipient is not exactly equal to the previous `BlockQuery's Block Counter, plus 1 (i.e. next block is being directly asked), a block counter SHALL be considered out-of-order by the recipient.

Blocks (**Block**, **BlockEOF** message) SHALL be sent in ascending and sequential Block Counter order. If the arriving Block Counter at the recipient is not exactly equal to current expected Block Counter, the block counter SHALL be considered out-of-order by the recipient.

Block acknowledgements (**BlockAck**, **BlockAckEOF** messages) SHALL be sent with the same Block Counter as the previously received **Block/BlockEOF**, since they convey a direct acknowledgement. If the arriving Block Counter at the recipient is not exactly equal to the last sent Block Counter, the Block Counter SHALL be considered out-of-order by the recipient.

On any out-of-order block counter condition described above, the recipient of the out-of-order message SHALL send a **StatusReport(GeneralCode: FAILURE, ProtocolId: BDX, ProtocolCode: BAD_BLOCK_COUNTER)** and abort the transfer. On receiving a **StatusReport(GeneralCode: FAILURE, ProtocolId: BDX, ProtocolCode: BAD_BLOCK_COUNTER)**, the recipient SHALL abort the transfer.

Since Block Counter fields are always 32-bit unsigned integer values, but file sizes may be specified